ROHIT RAKSHAN PALANIKUMAR

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EDUCATION

The University of Texas at Austin – Austin, TX **December 2026**
Bachelor of Science in Aerospace Engineering (Honors)
Computational Science and Engineering Certificate, GPA: 3.93
Electives: Rocket Science, Low Earth Orbit, Computational Methods in Structural Analysis
Activities: Engineering Honors, Longhorn Racing, AIAA, Longhorn Rocketry Association, ASE/EM Tutor

EXPERIENCE

The Boring Company – *Integration Engineering Intern* **September 2025 – December 2025**
- Supported assembly and integration of 3+ Prufrock TBMs across multi-disciplinary builds
- Managed MBOMs across 5 active projects with 500+ parts; corrected design errors to reduce integration rework
- Procured components and tracked inventory to sustain uninterrupted concurrent TBM assembly
- Partnered with production techs and test engineers to build and validate mechanical assemblies hands-on

Flamefront Propulsion – *Engineering Intern* **June 2025 – August 2025**
- Built turbojet engine modeling tool in Python from scratch using GitHub libraries, ingesting Mach, altitude, and geometry inputs to output thrust, SFC, and required propulsion performance metrics across the flight envelope
- Implemented component selection logic with mechanical and thermodynamic constraints; achieved 6% thrust increase
- Benchmarked model against experimental data, validating key performance parameters within 10% accuracy
- Designed engine geometry in CAD from component analysis outputs, closing analysis-to-hardware loop

Computational Fluid Physics Laboratory – *Undergraduate Researcher* **Oct 2024 – August 2025**
- Investigated programmable microjet flow control in turbulent boundary layers to extend attached flow over NACA 4412
- Ran high-fidelity CFD cases on TACC supercomputer using Nek5000; validated Cl/Cd against XFOIL achieving 93% prediction match via span increase
- Microjet made a 42% decrease in separation volume for fast LSMs; indicating aerodynamic performance improvement
- Generated structured meshes in GridPro and GMSH with constrained boundary layer tolerances; analyzed in ParaView

ORGANIZATIONS

Longhorn Racing Electric (FSAE Team) – *Aerodynamics Engineer* **August 2023 – Present**
- Designed flip-ups and bargeboard in SolidWorks; generating 40N downforce and 16N drag
- Conducted gurney flap study achieving 20% downforce increase and 7% drag increase
- Optimized sidepod geometry to hit radiator cooling inlet flow targets via ANSYS and Cadence CFD
- Manufactured aero components using carbon fiber layups and resin infusion; reinforced with aluminum ribs and structural spars

Texas Spacecraft Laboratory
Communications Test Engineer **February 2024 – May 2025**
- Conducted thermal-vacuum testing of spacecraft electronics across -50°C to +100°C and 10^{-3} Pa, documenting results and validating component performance under flight-representative environmental conditions
- Performed trade studies of S-Band ground station components to meet mission link requirements for 10cm CubeSat supporting NASA ARTEMIS GPS positioning

Lead Design Engineer
- Led FEA-based structural optimization of 1.25kg sensor payload (8×8×5 in) under 10N load case, ensuring structural integrity and integration readiness across subsystems
- Coordinated payload design decisions across subsystems to meet mission packaging requirements for a High Altitude Weather Balloon

LRA Vertical Active Stabilization – *Avionics Bay Sub-Team Lead* **August 2023 – March 2024**
- Led design and testing of fin geometries for active vertical stabilization, iterating configurations in SolidWorks and validating stability margins in OpenRocket
- Collaboratively designed and built avionics bay to house flight electronics, ensuring structural fit and component accessibility for integration and recovery

PROJECTS

TLOFT DARPA Heavy Lift Challenge

- Designed VTOL airframe CAD targeting 160 lb payload capacity on a 40 lb frame (4:1 lift-to-weight ratio) across a 7 ft wingspan and 39×39 in quadcopter frame
- Performed Euler buckling analysis to validate structural integrity at FOS of 5 for primary load-bearing members
- Simulated VTOL trajectory via rotor dynamics modeling, resolving optimal cruise AoA of 14° at 18 m/s for heavy-lift flight envelope

Automatic Headrest

- Designed a cost-effective off-the-shelf part that adjusts a car seat headrest using a linear actuator and switches (soldering)
- Integrated hidden wiring and bolted components onto the frame for maximum stability

SKILLS

- **Digital:** SolidWorks, Inventor, ANSYS, Cadence, MATLAB, ParaView , Java, C++, Python, R, HTML, GridPro, PyCycle, Git
- **Manufacturing:** Carbon Fiber Layups, CAM, Mills, Lathe, 3D Printing, Soldering, Sanding, Coating, Resin Infusion
- **Other:** System Assembly Integration & Testing, Thermo Simulations, SW Dev, Team Management, FEA, CFD